

Project Statement

Problem Statement

There is a need for lightweight, terminal-based applications that help students build problem-solving and logical deduction skills without the distraction of complex graphical interfaces. Many beginners struggle to find simple, progressive challenges that test both their technical vocabulary and deductive reasoning.

Scope of the Project

The "Guess the Mystery Word" project is a console-based Python application designed to challenge users through a series of progressive riddles. The scope includes a 3-level game loop, string manipulation for formatting and input validation, and a state-tracking system for user attempts.

Target Users

- Computer Science students looking to reinforce technical vocabulary (e.g., Algorithms, Variables).
- Individuals seeking quick, terminal-based cognitive exercises.

High-level Features

- **Progressive Level System:** Three distinct levels with unique riddles and decreasing margin for error.
- **Terminal UI Formatting:** Utilizes ANSI escape sequences for bold and italicized text to enhance readability.
- **Input Normalization:** Automatically handles user input variations using string capitalization methods.
- **Dynamic Attempt Tracking:** Provides real-time feedback on remaining chances.